10/4/26

Tutorial - 12

Bootstrap → Sampling with Replacement

Sample:	feature 1 (Age)	feature 2 (income)	feature 3 (credit score)	Target (loan Approved)
S1	25	40	650	Yes
S2	30	60	700	Yes
S3	45	33	600	NO
S4	50	30	580	NO
S5	28	55	720	Yes

TASKS:-

1. create 3 bootstrap samples (size = 5) from the dataset
2. Train 3 decision tree models using each bootstrap sample.
3. for a new sample (Age = 32, Income = 50, credit score = 690) determine predictions from all trees
4. compute the final prediction using majority voting.

1) Bootstrap 1 (B1): S1, S2, S2, S4, S5

Bootstrap 2 (B2): S2, S3, S4, S4, S5

Bootstrap 3 (B3): S1, S2, S3, S3, S4

2) Decision Tree(B1) | Bootstrap 2(B2) | Bootstrap 3(B3)

B1	B2	B3	B4	B5	B2	B3	B4	B4	B5	B1	B2	B3	B3	B4
Y	Y	Y	N	Y	Y	N	N	N	Y	Y	Y	N	N	N

Age: ≤30 → Yes; >30 → NO

credit score: ≥650 → Yes; <650 → NO

Income: >40 → Yes; <40 → NO

single split DT

34 4) prediction & voting

Age = 32 → loan Approval : N

Income = 50 → loan Approval : Y

credit score = 690 → Y

majority voting

N Y Y (Y)

Age = 35 → loan Approval : N

Income = 45 → Y

credit score = 750 → Y

majority voting

N Y Y (Y)